Fig. 4: Light flicker analyser (Image courtesy: www.everfine.net)

manufacturers but also customers.

The minimum diameter of the sphere available is 8cm, while the maximum diameter is up to 3 metres. The 8cm spheres are used by LED chip manufacturers to test the lumens of a single LED chip or an array, while sizes above 8cm are used to test the lumens output of LED luminaires.

Image sensor calibration is the utmost important feature that has been implemented in the integrating sphere equipment recently. It helps in producing good-quality images by fixing tiny defects both in the optical system and camera sensors.

Software upgraded in goniophotometer

A goniophotometer is used to measure the luminous intensity and luminous intensity distribution of LED luminaires. It is available in type A, B and C, of which type B has its burning position moving. But now the burning position or the location where luminaire is kept has been fixed.

Goniophotometer is a complete and sophisticated instrument in itself. It hardly requires any hardware upgrades. But its updated software features new colour evaluation.

Near-field goniophotometer. Besides the intensity and spatial luminous distribution test of LED luminaires, the LED itself is also tested by manufacturers using a near-field goniophotometer—one of the test equipment released lately.

Thermal resistance tester for LED chips

Junction temperature is the most important factor that decides the reliability and life-time of an LED. A thermal resistance tester can be used for a single LED chip or LED arrays. This equipment works on the concept of an LED's forward voltage. It gives a large current input to the LED in order to achieve steady and transition thermal resistance measurement.

Light flicker analyser

Usually, LED luminaire flickering is caused by a fault in the driver, poor environmental conditions, or temperature issue with the power supply and LED load. Test equipment manufacturers have come up with solutions that detect flicker and show safe limit conditions.

Light flicker analyser is a portable instrument that gives flicker index, percentage flicker and fundamental frequency. So users can easily make out whether the slightest of flicker in the LED product is under the safe category as per standards for flicker hazard. The instrument is equipped with a high-speed photometer head to